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Squash training system

Abstract

A squash training system includes a sport sensor. The sport sensor includes a sensing unit configured to detect a motion of a racquet and generate three-dimensional (3D) motion data associated with the motion. The sport sensor also includes a memory configured to store the 3D motion data and a communication interface configured to transmit the 3D motion data to a terminal device. The squash system also includes at least one processor configured to receive the 3D motion data from the sport sensor and determine, based on the 3D motion data, at least one characteristic associated with the motion. The processor is also configured to render a graphical representation of the characteristic on a display.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5114150	12/1991	Matsumura	N/A	N/A
8840483	12/2013	Steusloff et al.	N/A	N/A
2004/0185962	12/2003	Yabu	473/345	A63B 60/46
2008/0085778	12/2007	Dugan	N/A	N/A
2008/0108462	12/2007	Krysiak	N/A	N/A
2011/0183787	12/2010	Schwenger et al.	N/A	N/A
2012/0157241	12/2011	Nomura et al.	N/A	N/A
2014/0018181	12/2013	Blake	473/409	G09B 19/0038
2014/0277630	12/2013	Meadows	700/91	G09B 19/0038
2015/0051009	12/2014	Davenport	N/A	N/A
2015/0119159	12/2014	Sato et al.	N/A	N/A
2015/0141178	12/2014	Mettler	473/553	A63B 60/46
2015/0335978	12/2014	Syed	473/223	A63B 24/0006
2015/0375080	12/2014	Boisvert	473/239	A63B 69/3632
2017/0312609	12/2016	Thornton et al.	N/A	N/A

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Background/Summary

RELATED APPLICATION (1) This application is a division of U.S. patent application Ser. No. 16/251,028, filed Jan. 17, 2019, which claims the benefits of priority to U.S. Provisional Application No. 62/618,592, filed Jan. 17, 2018. The entire contents of each of the above-quoted applications are expressly incorporated herein by reference.

TECHNICAL FIELD

(1) The present application relates to squash sport, a squash training system, and a method for assisting squash training. More particularly, the present application relates to a squash training system utilizing a sport sensor to detect motions of a racquet in practicing squash.

BACKGROUND

(2) Squash is an indoor racquet sport. The game of squash is played by two players (or four players for doubles) with racquets in a four-walled court with a small, hollow rubber ball. A player can direct the direction and control the speed of the ball as the racquet (or strings on the racquet) impacts the ball with varying angles and force. Traditionally, squash training relies on an on-site coach, who provides instructions to the players based on observation and experience. However, hiring a coach is expensive, and the availability of the coach may not be guaranteed. In addition, this instruction-based training approach lacks quantitative measures to evaluate how well a particular action, such as a swing, is performed. Some players make video recordings of their training routines and analyze their swings afterwards. While this approach provides a basis (e.g., the video recordings) for finer analyses, it lacks real time feedback to allow the players to make immediate improvements.

(3) Embodiments of the present disclosure address the aforementioned problems by a squash training system providing instant feedback with rich quantitative performance measures based on racquet motion data collected by a sport sensor.

SUMMARY

(4) In one aspect, a sport sensor for sensing motions of a racquet is provided. The sport sensor may include at least one sensing unit configured to detect a motion of the racquet and generate three-dimensional (3D) motion data associated with the motion. The sport sensor may also include a memory configured to store the 3D motion data and a communication interface configured to transmit the 3D motion data to a terminal device.

(5) In another aspect, a squash training system is provided. The system may include a sport sensor associated with a racquet. The sport sensor may be configured to detect a motion of the racquet and generate 3D motion data associated with the motion. The system may also include at least one processor. The processor may be configured to receive the 3D motion data from the sport sensor. The processor may also be configured to determine, based on the 3D motion data, at least one characteristic associated with the motion. The processor may be further configured to render a graphical representation of the characteristic on a display.

(6) In another aspect, a method for assisting squash training is provided. The method may include detecting, by a sport sensor, a motion of a racquet. The method may also include generating 3D motion data associated with the motion. The method may further include determining, based on the 3D motion data, at least one characteristic associated with the motion. In addition, the method may include rendering a graphical representation of the characteristic on a display.

(7) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a schematic diagram of an exemplary squash training system, according to embodiments of the disclosure.

(2) FIG. 2 illustrates a schematic diagram of an exemplary sport sensor, according to embodiments of the disclosure.

(3) FIG. 3 illustrates a block diagram of a terminal device used in the system shown in FIG. 1, according to embodiments of the disclosure.

(4) FIG. 4 illustrates an exemplary motion profile, according to embodiments of the disclosure.

(5) FIG. 5 is a flowchart of an exemplary method for assisting squash training, according to embodiments of the disclosure.

(6) FIGS. 6, 7A-7F, and 8-10 are exemplary user interfaces displayed by the terminal device shown in FIG. 3, according to embodiments of the disclosure.

DETAILED DESCRIPTION

(7) Reference will now be made in detail to the exemplary embodiments, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(8) FIG. 1 illustrates an exemplary squash training system **100**, according to embodiments of the disclosure. System **100** may include a sport sensor **110** associated with a racquet **102**. Racquet **102** may be a squash racquet used by a player to hit a squash ball during practicing squash. As used herein, the term “squash training” includes any activities of playing, practicing, or learning squash, including, for example, a squash training session, a squash practicing session, a squash learning session, a squash game, a squash match, a squash competition, etc. As shown in FIG. 1, sport sensor **110** may be attached to, mounted on, or otherwise associated with racquet **102** in various locations and in differing ways. For example, sport sensor **110** may be attached to (e.g., by an elastic pouch, a sticky pad, one or more screws, or any combination thereof) the base of the racquet handle. In another example, sport sensor **110** may be attached to the neck or the head of racquet **102**. In a further example, sport sensor **110** may be attached to the strings of racquet **102**. In a further example, sport sensor **110** may be embedded into racquet **102**, for example, placed within the handle, neck, or head portion of racquet **102**. The manner in which sport sensor **110** is attached to racquet **102** can vary depending on particular applications.

(9) In some embodiments, sport sensor **110** may be removably attached to racquet **102** such that sport sensor **110** can be detached from racquet **102** after being attached to racquet **102**. For example, sport sensor **110** may be held within a silicon sleeve, which in turn can be attached to as well as detached from the base of racquet **102**. In another example, sport sensor **110** may be attached to the body of racquet **102** using a sticky tape, which may be removed without substantially damaging the racquet and/or sport sensor **110**. In this way, multiple sport sensors can be switched (e.g., in case any one of them is out of power) without interrupting the squash training.

(10) System **100** may also include a terminal device **120**. Terminal device **120** may include, for example, a smart phone, a wearable device (e.g., a smart watch), a hand-held device, a tablet, a laptop, etc. Terminal device **120** may wirelessly connect to sport sensor **110** to exchange information with sport sensor **110**. After receiving information collected by sport sensor **110**, terminal device may process the information and, based on the information, render graphical representations on a display of terminal device **120**.

(11) In some embodiments, terminal device **120** may also exchange information with a server **130**. Server **130** may include any suitable type of servers, including a local server, a remote server, a cloud server, a distributed server, a physical server, a virtual server, or any other suitable computing system. Server **130** may store information related to squash training. For example, information received or generated by terminal device **120** may be stored in sever **130**. Server **130** may also perform various functions disclosed herein in conjunction with terminal device **120**.

(12) FIG. 2 illustrates a schematic diagram of an exemplary implementation of sport sensor **110**, according to embodiments of the disclosure. The upper part of FIG. 2 shows a side view of sport sensor **110**, while the lower part of FIG. 2 shows a top view of sport sensor **110**. As shown in FIG. 2, sport sensor **110** may include a crown portion **220** and a base portion **210**. Crown portion **220** may include a power switch **230** on an upper surface of sport sensor **110**. A user may press power switch **230** to turn on/off sport sensor **110**. The upper surface may exhibit a design pattern, such as polygons **222**. Other design patterns may also be used.

(13) Base portion **210** may host one or more hardware components of sport sensor **110**, including,

for example, a power source **212**, one or more sensing units **214**, a communication interface **216**, a processor **218**, and a memory **219**. Power source **212** may be a battery (e.g., an alkaline battery, a button cell battery, a lithium-ion battery, etc.), an induction coil (e.g., for providing power through induction coupling), an RF antenna (e.g., for providing power through RF coupling), or any source of electric power. Power source **212** may supply power to other components of sport sensor **110**.

(14) Sensing unit(s) **214** may include a motion sensing unit (e.g., a linear motion sensing unit, an angular motion sensing unit, etc.), a direction sensing unit, an orientation sensing unit, an impact sensing unit, or any other sensing units for detecting motions, force, and/or power. For example, a motion sensing unit can be implemented by an accelerometer, a gyroscope, a combination thereof, etc.

(15) Consistent with the embodiments disclosed herein, sensing unit(s) **214** may detect a motion of racquet **102** and generate 3D motion data. For example, the motion may be a swing of racquet **102**. Sensing unit **214** may detect that racquet **102** moves during the swing and generate, for example, 3D linear and/or angular motion data corresponding to the swing. In some embodiments, the 3D linear motion data may include data representing linear displacement, speed, acceleration, and/or other parameters in three orthogonal directions. In some embodiments, the 3D angular motion data may include data representing angular displacement, speed, acceleration, and/or other parameters in three orthogonal directions. In some embodiments, sensing unit(s) **214** may also detect the speed of racquet **102** (e.g., at the head of racquet **102**) and generate the speed data. In some embodiments, when racquet **102** or strings of racquet **102** hit a squash ball, the impact between the squash ball and racquet **102** or between the squash ball and the strings of racquet **102** may be detected and impact data may be generated by sensing unit(s) **214**. The impact data may include data representing the force, power, time of impact, speed of impact, or other parameters relating to the impact. The 3D motion/impact data may be transmitted to terminal device **120** for further processing.

(16) Communication interface **216** may include one or more communication adaptors to transmit and/or receive information with an external device, such as terminal device **120**. In some embodiments, communication interface **216** may include a short-range wireless communication adaptor such as a Bluetooth™ module, an NFC module, etc. In some embodiments, communication interface **216** may include a wireless local area network (WLAN) adaptor. In some embodiments, communication interface **216** may include a wired communication adaptor such as a USB port. Sport sensor **110** may establish a communication link with terminal device **120** through communication interface **216** to transmit, for example, 3D motion data and impact data.

(17) Processor **218** may include any suitable type of microprocessor, microchip, microcontroller, etc. Processor **218** may be configured to control other components of sport sensor **110**, such as sensing unit(s) **214** to detect motion/impact and/or generate data. Processor **218** may also pre-process the generated data. In some embodiments, processor **218** may control establishment of a communication link between sport sensor **110** and terminal device **120** and transmission of data from sport sensor **110** to terminal device **120**.

(18) Memory **219** may include any suitable type of memory device, such as a RAM, a ROM, a flash drive, a memory card, etc. Memory **219** may store instructions executable by processor **218** to perform various functions of processor **218**. Memory **219** may also store data (e.g., 3D motion data and/or impact data) generated by sensing unit(s) **214**.

(19) FIG. 3 shows a block diagram of an exemplary implementation of terminal device **120**. As shown in FIG. 3, terminal device **120** may include a processor **340**, a memory **330**, a display device **350**, and a communication interface **320**. Terminal device **120** may implement various functions disclosed herein alone or in conjunction with sport sensor **110** and/or server **130**.

(20) Communication interface **320** may be configured to send information to and receive information from other components of system **100** via communication links indicated by arrowed lines shown in FIG. 1. In some embodiments, communication interface **320** can include an

integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection. As another example, communication interface **320** can include a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links can also be implemented by communication interface **320**. In such an implementation, communication interface **320** can send and receive electrical, electromagnetic or optical signals that carry digital data streams representing various types of information via the communication links.

(21) Processor **340** may include one or more processing devices configured to perform functions of the disclosed embodiments. Processor **340** may include any appropriate type of general-purpose or special-purpose microprocessor, digital signal processor, graphic processor, or microcontroller. In some embodiments, processor **340** may include a single core or multiple cores executing parallel processes simultaneously. For example, processor **340** can be a single-core processor configured with virtual processing technologies. In certain embodiments, processor **340** uses logical processors to simultaneously execute and control multiple processes. Processor **340** can implement virtual machine technologies, or other known technologies to provide the ability to execute, control, run, manipulate, and store multiple software processes, applications, programs, etc. In another embodiment, processor **340** may include a multiple-core processor arrangement (e.g., dual core, quad core, etc.) configured to provide parallel processing functionalities that allow processor **340** to execute multiple processes simultaneously. It is appreciated that other types of processor arrangements can be implemented that provide for the capabilities disclosed herein.

(22) Memory **330** may include a volatile or non-volatile, magnetic, semiconductor-based, tape-based, optical, removable, non-removable, or other type of storage device or tangible and/or non-transitory computer-readable medium that stores one or more executable programs, such as a squash training application and an operating system. The programs may include software that, when executed by processor **340**, performs various functions or operations to be discussed in greater detail below. The programs may also include communication software that, when executed by processor **340**, provides communications using communication interface **320**, such as communicating with sport sensor **110** and/or server **130**.

(23) Display **350** may include a display such as a Liquid Crystal Display (LCD), a Light Emitting Diode Display (LED), a plasma display, or any other type of display, and provide a Graphical User Interface (GUI) presented on the display for user input and data depiction. The display may include a number of different types of materials, such as plastic or glass, and may be touch-sensitive to receive inputs from the user. For example, the display may include a touch-sensitive material that is substantially rigid, such as Gorilla Glass™, or substantially pliable, such as Willow Glass™.

(24) In some embodiments, sport sensor **110** may detect motions and/or impacts, generate 3D motion data and/or impact data, and transmit the generated data to terminal device **120**. For example, sport sensor **110** may connect with terminal device **120** (e.g., via a wireless communication link) and transmit data in real time. In another example, sport sensor **110** may transmit data to terminal device **120** upon query from terminal device **120**. In this way, data may be transmitted to terminal device periodically or in a lump-sum manner. For instance, data generated by generated by sensing unit(s) **214** and stored in memory **219** during a training session. After the training session concludes, sport sensor **110** may be connected to terminal device **120** and transmit the data stored in memory **219** to terminal device **120**.

(25) After receiving data from sport sensor **110**, processor **340** may analyze the data and determine performance metrics. For example, processor **340** may determine, based on the received 3D motion data, one or more characteristics associated with a motion corresponding to the 3D motion data. In some embodiment, the motion may correspond to a swing of racquet **102**. A swing may refer to a motion of racquet **102** starting from a relatively stable state to an acceleration state, followed by a deceleration state, and return to a relatively stable state.

(26) In some embodiments, sport sensor **110** (e.g., using processor **218**) may distinguish individual

or unique swings. For example, when a player swings racquet **102** multiple times, sport sensor **110** may detect an individual swing out of the multiple swings. In some embodiments, sport sensor **110** may determine an individual swing by detecting the acceleration/deceleration states of racquet **102**. In some embodiments, sport sensor **110** may determine an individual swing by detecting the speed of racquet **102** (e.g., for a relatively low speed to a relatively high speed and back to a relatively low speed). In some embodiments, sport sensor **110** may determine an individual swing by detecting the displacement of racquet **102** (e.g., linear displacement and/or angular displacement from a reference point). In some embodiments, sport sensor **110** may also apply filtering, denoising, smoothing, or other signal processing techniques to improve the signal-to-noise ratio before determining individual swings. In some embodiments, sport sensor **110** may use threshold values (e.g., acceleration, speed, displacement, etc.) to determine the starting/ending points of an individual swing.

(27) In some embodiments, determination of individual swings may be performed by processor **340** or jointly by processors **218** and **340**. For example, sport sensor **110** may transmit raw data to terminal device **120** without preprocessing, and terminal device **120** may, using processor **340**, to analyze the raw data and determine individual swings. In another example, sport sensor **110** may pre-process the raw data (e.g., denoising, filtering, smoothing, thresholding, etc.) before transmitting the data to terminal device **120**. Terminal device **120** may then analyze the pre-processed data to determine individual swings.

(28) In some embodiments, the type of an individual swing may be determined by processor **340** based on the received 3D motion data. For example, a squash swing can be categorized as a forehand swing or a backhand swing. In another example, a squash swing can be categorized as a drive, a lob, a drop, a boast, etc., depending on, for example, the way racquet **102** moves, the angle of racquet **102**, the timing of impact (e.g., early, in-line, or late), etc. Different types of swings have different characteristics in terms of the 3D motion data associated therewith. Processor **340** may be configured to recognize the type of an individual swing based on the 3D motion data.

(29) In some embodiments, processor **340** may determine the type of an individual swing based on a motion profile extracted from the 3D motion data. FIG. 4 illustrates an exemplary motion profile **400**. Referring to FIG. 4, motion profile **400** may be represented by the values of angular displacement as a function of time, shown in FIG. 4 as curve **410**. The angular displacement data may be extracted from the 3D motion data received from sport sensor **110**. Curve **410** exhibits certain features, such as a relatively flat starting portion, following by a substantially linear increase and a substantially linear decrease, and eventually back to the substantially same level as the starting portion. These features, including the shape of curve **410**, the timing of the increase/decrease, the slope of the increase/decrease, the relative levels between the highest and lowest points, etc., may be used to characterize a particular type of swing. For example, curve **410** may correspond to a forehand drive swing. Processor **340** may compare the motion profile extracted from the 3D motion data with a reference motion profile characterizing a certain type of swing to determine the type of the motion (e.g., swing) corresponding to the motion profile. For example, one or more reference motion profiles may be stored in memory **330** and processor **340** may compare an extracted motion profile with the reference motion profiles and determine the type of the swing to be the one exhibiting the closest match. In another example, memory **330** may store certain features of the reference motion profiles instead of the actual curve and processor **340** may first calculate the corresponding features based on the 3D motion data for an individual swing and determine the type of the swing based on comparing the features. In some embodiments, the features may be stored in a form of a series of fundamental waveforms that collectively form or approximate the reference motion profile (e.g., via Fourier transform, spectrum analysis, etc.). The comparison between the extracted motion profile and the reference motion profile may be performed by comparing their corresponding fundamental waveforms.

(30) In some embodiments, processor **340** may receive impact data from sport sensor **110** and

determine an impact metric based on the impact data. For example, the impact data may include the force applied to racquet **102** or the strings of racquet **102**, the time of impact, the relative speed of the impact, the acceleration of racquet **102**, etc. Based on the impact data, processor **340** may determine power or force associated with the impact. For example, if the speed v of racquet **102** and the impact force F is detected by sensor **110**, then the power P of the impact can be calculated as $P=F \times v$. In another example, the force F of the impact can be calculated as $F=m \times a$, where a is the acceleration of racquet **102** and m is the mass of the squash ball (e.g., a known value) hitting the strings of racquet **102**. Other methods may also be used to determine the power or force of the impact.

(31) FIG. 5 is a flowchart of an exemplary method **500** for assisting squash training, according to embodiments of the disclosure. Method **500** may be implemented by system **100** that includes, among other things, sport sensor **110**, memory **330**, and processor **340** that performs various operations disclosed herein. In some embodiments, method **500** may be jointly performed by sensor **110**, terminal device **120**, and/or server **130**. It is to be appreciated that some of the steps of method **500** may be optional to perform the disclosure provided herein, and that some steps may be inserted in the flowchart of method **500** that are consistent with other embodiments according to the current disclosure. Further, some of the steps may be performed simultaneously, or in an order different from that shown in FIG. 5.

(32) In step **510**, sport sensor **110** may detect, using sensing unit(s) **214**, a motion of racquet **102**. For example, the motion may correspond to an individual swing and sport sensor **110** may distinguish the individual swing from other swings. In step **520**, sport sensor **110** may generate 3D motion data and/or impact data. For example, the 3D motion data may include linear and/or angular motion data in three orthogonal directions. The impact data may include force, time of impact, speed, or other relevant data associated with the impact. The generated data may be stored in memory **219** and may be transmitted to terminal device **120** through communication interface **216**.

(33) In step **530**, processor **340** may determine one or more characteristics associated with the motion. For example, processor **340** may determine the type of a swing based on the associated motion profile extracted from the 3D motion data. In some embodiments, processor **340** may determine whether a swing is a forehand swing or a backhand swing. Processor **340** may further determine whether a swing is a drive, volley, drop, cross-court, boast, or lob swing, depending on the motion of racquet **102** (e.g., linear and/or angular profile as a function of time), the speed of racquet **102**, the force/power of impact, etc. In some embodiments, processor **340** may determine whether a swing is a serve swing and/or whether the serve swing is a hard or lob serve swing. In some embodiments, processor **340** may determine the power and/or force of an impact between racquet **102** and a squash ball. Processor **340** may determine whether a swing hits a sweet spot that would optimize the performance based on a reference metric or a predetermined criterion. For example, a sweet spot for hitting the squash ball in a forehand drive may require the swing power, swing speed, and/or swing time to be within a certain range. Processor **340** may compare the metrics determined from the 3D motion data and/or impact data with reference metrics.

(34) In step **540**, processor **340** may render one or more graphical representations of the determined characteristic(s) on display device **350**. FIGS. 6, 7A-7F, and 8-10 illustrate exemplary interfaces containing exemplary graphical representations. These exemplary interfaces may be implemented by a squash training application (e.g., a mobile app) running on terminal device **120**.

(35) FIG. 6 shows an exemplary main interface **600**. Referring to FIG. 6, main interface **600** may include a chart **610** indicating the total number of swings and the percentage numbers of each type of swings. For example, chart **610** shows that out of the 100 swings, 10% are drop swings, 30% are lob swings, 20% are drive swings, and 40% are other types of swings (e.g., swings that are not within any of the drop, lob, and drive categories). Main interface **600** also include an information bar **620** indicating the percentage numbers of the forehand and backhand swings. For example, information bar **520** shows that the 100 swings are all backhand swings. A detail information

section **630** shows the number of swings of each type (drop, lob, drive, other) and the percentage numbers of forehand and backhand swings within each type. Button **640** is used to open another interface showing comparisons between the player's performance and a professional's performance. Button **650** is used to open another interface for recording a new training session. A tab section **660** is provided at the bottom of main interface **600** showing other functional tabs provided by the squash training application.

(36) FIGS. 7A-7F show exemplary comparison interfaces, which may be displayed after button **640** is clicked. Referring to FIG. 7A, a comparison interface **700** includes a swing type indication bar **710**, a swing assessment section **720**, a swing curve comparison section **740**, and a swing data comparison section **750**. Swing type indication bar **710** shows the type of swing currently being compared. For example, swing type indication bar **710** shows that forehand drive is the type of swing currently being compared with data from professional(s)/coach(s). Swing assessment section **720** includes a percentage indicator indicating the degree of matching, and individual ratings focusing on various aspects. For example, a three-star rating system is used to assess the play's forehand drive performance in the pilot stage, force phase, shoot stage, as well as the shot speed of the swing. Based on the analysis of the 3D motion data and/or impact data generated by sport sensor **110**, performance data (e.g., in various stages/phase) may be determined by processor **340** (e.g., based on the time or features of the motion curve). These performance data may be compared with corresponding performance data obtained from professional players or coaches to evaluate the performance of the player. For example, swing curve comparison section **740** shows a curve of a professional play (solid line) and a corresponding curve (dashed line) of the player. In addition, swing data comparison section **750** may show comparisons of performance metrics such as swing speed, swing power, and swing time.

(37) FIGS. 7B-7F show similar interfaces but with different types of swings. FIG. 7B shows a comparison interface of backhand drive swings and a professional curve **741**. FIG. 7C shows a comparison interface of forehand lob swings and a professional curve **742**. FIG. 7D shows a comparison interface of backhand lob swings and a professional curve **743**. FIG. 7E shows a comparison interface of forehand drop swings and a professional curve **744**. FIG. 7F shows a comparison interface of backhand drop swings and a professional curve **745**. In some embodiments, the professional curves may also be used as reference curves to determine the type of an individual swing.

(38) FIG. 8 shows an interface **800** of recording a training session or a game. Interface **800** may be activated by clicking the "Play" tab in section **660** of the main interface **600**. The player can start the recording by clicking the "START" button displayed on interface **800**. The play time, number of swings, the fastest swing, and calories may be also displayed on interface **800**. During a game play, the power, racquet head speed, and/or swing speed values can be displayed on interface **800** and may be refreshed for individual swings. For example, when racquet **102** impacts a squash ball, the value(s) of the power, racquet head speed, and/or swing speed may be displayed or refreshed to reflect the instant impact metrics. FIG. 8 also includes a video recording button **810** to allow the player to video record the practice or training session. The recorded video may be stored locally on terminal device **120** (e.g., in memory **330**) and/or may be uploaded to server **130**. Video analysis may be performed in conjunction with the motion/impact analysis to determine performance metrics. In some embodiments, videos of professional players and/or coaches may be used to compare with the video of the player.

(39) FIG. 9 shows a real-time 3D interface **900**. Interface **900** may be displayed after clicking the "REAL-TIME" tab in tab section **660** of interface **600**. Interface **900** may display the motion of racquet **102** in a 3D setting based on 3D motion data generated by sport sensor **110**. In some embodiments, the 3D display may be a replay of a motion captured by sport sensor **110**. In some embodiments, the 3D display may be a real time display of the motion of racquet **102** as the player is playing the game. Performance metrics, such as power, speed, swing duration, may be displayed

on interface **900**.

(40) FIG. **10** shows a training interface **1000**. Interface **1000** may be displayed after clicking button **650** displayed on interface **600**. Interface **1000** may display the number of a particular type of swings the player practices. For example, as the player is practicing swings, the type of an individual swing can be determined by processor **340**. If the type matches the type shown in interface **1000**, then the number of total swings of that particular type may be incremented **1**. Otherwise, the number of the total swing of that particular type may stay the same.

(41) In some embodiments, a player may set up the profile of racquet **102** such that the performance metrics may be calibrated based on the particular profile. For example, the reference motion profiles used to determine the type of swings may be adjusted accordingly based on the calibration.

(42) In some embodiments, processor **340** may analyze the distribution of swings based on the types, impact metrics, or other performance metrics. For example, the number of a particular type of swing may be determined, the associated intensity, such as the power/speed/force of the swings may be analyzed.

(43) In some embodiments, a coach may be allowed to access the performance metrics of a student or trainee to evaluate the training progress. For example, terminal device **120** of an individual trainee may upload performance metrics to server **130**. The coach may then access the data stored in server **130** to access the performance metrics of the individual trainee.

(44) In some embodiments, a coach-version squash training application or a coach mode of the squash training application may be provided to display a coach portal interface. The coach portal interface may display one or more trainees' squash training performance metrics and may generate reports of the trainee users. The coach portal interface may be displayed on a terminal device similar to terminal device **120** (e.g., tablet, computer browser, smartphone, etc.)

(45) In some embodiments, the coach portal interface may allow a coach to view his/her students/trainees via an opt-in process or through a “friend” function as a security layer. In some embodiments, the coach portal interface may provide review functions to allow the coach to review student user data. The coach portal interface may also provide reporting functions to generate reports from an individual student user data or an aggregate of all students of the coach according to sortable values such as swing speed, racquet head speed, and/or power.

(46) In some embodiment, the coach portal interface may provide an exporting function to export screen displays, user data, analyses, reports, or the like to an electronic file (e.g., PDF file) for sharing (e.g., via email) or printing out.

(47) In some embodiments, the coach portal interface may contain similar elements to interfaces **600**, **700**, **800**, **900**, and **1000**, with additions and/or adjustments designed specifically for coaches.

(48) Another aspect of the disclosure is directed to a non-transitory computer-readable medium storing instructions which, when executed, cause one or more processors to perform the methods, as discussed above. The computer-readable medium may include volatile or non-volatile, magnetic, semiconductor-based, tape-based, optical, removable, non-removable, or other types of computer-readable medium or computer-readable storage devices. For example, the computer-readable medium may be the storage device or the memory module having the computer instructions stored thereon, as disclosed. The computer-readable medium may be a disc, a flash drive, or a solid-state drive having the computer instructions stored thereon.

(49) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed system and related methods. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the disclosed system and related methods.

(50) It is intended that the specification and examples be considered as exemplary only, with a true scope being indicated by the following claims and their equivalents.

Claims

1. A method for assisting squash training, comprising: detecting, by a sport sensor, a motion of a racquet; generating motion data associated with the motion, wherein the motion data comprise a motion profile in a form of a first series of spectral fundamental waveforms obtained via spectrum analysis; comparing the motion profile with a plurality of reference profiles to determine a target reference profile that has a closest match to the motion profile at least by: comparing a corresponding series of spectral fundamental waveforms of each reference profile with the first series of spectral fundamental waveforms of the motion profile, wherein each of the plurality of reference profiles is associated with a different category of swing, and the corresponding series of spectral fundamental waveforms of different reference profiles are different from one another; and determining a type of swing associated with the motion from the different categories of swings based at least in part on the target reference profile, wherein the method further comprises: detecting, by the sport sensor, an impact between a squash ball and the racquet or between the squash ball and strings of the racquet; generating impact data associated with the impact; and determining an impact metric comprising power associated with the impact based on the impact data, wherein the power associated with the impact is determined based on a speed of the racquet and an impact force detected by the sport sensor.
2. The method of claim 1, wherein the motion data comprise linear motion data.
3. The method of claim 1, wherein the motion data comprise angular motion data.
4. The method of claim 1, wherein: detecting the motion of the racquet comprises detecting the speed of the racquet; and the method further comprises determining that the motion corresponds to an individual swing of the racquet when the detected speed changes from a relatively low speed to a relatively high speed and then changes back to another relatively low speed.
5. The method of claim 1, wherein: detecting the motion of the racquet comprises detecting a displacement of the racquet; and the method further comprises determining that the motion corresponds to an individual swing of the racquet when the detected displacement exceeds a threshold from a reference point.
6. The method of claim 1, wherein: the type of swing comprises at least one of a serve swing, a forehand swing, a backhand swing, a drive, a lob, a drop, or a boast.
7. The method of claim 1, comprising: rendering a real-time 3D display of the motion based at least in part on the type of swing.
8. The method of claim 1, comprising: rendering a comparison between the motion profile and at least one of the plurality of reference profiles.
9. A method for assisting squash training, comprising: receiving, from a sport sensor, motion data associated with a motion of a racquet detected by the sport sensor; determining a motion profile associated with the motion based on the received motion data, wherein the motion profile comprises a first series of spectral fundamental waveforms obtained via spectrum analysis; comparing the motion profile with a plurality of reference profiles to determine a target reference profile that has a closest match to the motion profile at least by: comparing a corresponding series of spectral fundamental waveforms of each reference profile with the first series of spectral fundamental waveforms of the motion profile, wherein each of the plurality of reference profiles is associated with a different category of swing, and the corresponding series of spectral fundamental waveforms of different reference profiles are different from one another; and determining a type of swing associated with the motion from the different categories of swings based at least in part on the target reference profile, wherein the method further comprises: detecting, by the sport sensor, an impact between a squash ball and the racquet or between the squash ball and strings of the racquet; generating impact data associated with the impact; and determining an impact metric comprising power associated with the impact based on the impact data, wherein the power

associated with the impact is determined based on a speed of the racquet and an impact force detected by the sport sensor.

10. The method of claim 9, wherein the motion data comprise linear motion data.

11. The method of claim 9, wherein the motion data comprise angular motion data.

12. The method of claim 9, wherein: the motion data comprise information of the speed of the racquet; and the method further comprises determining that the motion corresponds to an individual swing of the racquet when the speed changes from a relatively low speed to a relatively high speed and then changes back to another relatively low speed.

13. The method of claim 9, wherein: the motion data comprise information of a displacement of the racquet; and the method further comprises determining that the motion corresponds to an individual swing of the racquet when the displacement exceeds a threshold from a reference point.

14. The method of claim 9, wherein: the type of swing comprises at least one of a serve swing, a forehand swing, a backhand swing, a drive, a lob, a drop, or a boast.

15. The method of claim 9, comprising: rendering a graphical representation of the determined type of swing on a display.

16. The method of claim 9, comprising: rendering a real-time 3D display of the motion based at least in part on the type of swing.

17. The method of claim 9, comprising: rendering a comparison between the motion profile and at least one of the plurality of reference profiles.

18. A non-transitory computer-readable medium storing instructions that, when executed by at least one processor, cause the at least one processor to perform a method for assisting squash training, the method comprising: receiving, from a sport sensor, motion data associated with a motion of a racquet detected by the sport sensor; determining a motion profile associated with the motion based on the received motion data, wherein the motion profile comprises a first series of spectral fundamental waveforms obtained via spectrum analysis; comparing the motion profile with a plurality of reference profiles to determine a target reference profile that has a closest match to the motion profile at least by: comparing a corresponding series of spectral fundamental waveforms of each reference profile with the first series of spectral fundamental waveforms of the motion profile, wherein each of the plurality of reference profiles is associated with a different category of swing, and the corresponding series of spectral fundamental waveforms of different reference profiles are different from one another; and determining a type of swing associated with the motion from the different categories of swings based at least in part on the target reference profile, wherein the method further comprises: detecting, by the sport sensor, an impact between a squash ball and the racquet or between the squash ball and strings of the racquet; generating impact data associated with the impact; and determining an impact metric comprising power associated with the impact based on the impact data, wherein the power associated with the impact is determined based on a speed of the racquet and an impact force detected by the sport sensor.

19. The computer-readable medium of claim 18, wherein the motion data comprise linear motion data.

20. The computer-readable medium of claim 18, wherein the motion data comprise angular motion data.
